# Capital Markets

13 July 2020 17:51

| | |
|---|---|
| Capital Markets | Market for financial Instruments. |
| Participants | Issuers, investors, and intermediaries. |
| Primary Markets | • Securities markets<br>• Derivatives markets<br>• Currency markets<br>• Insurance markets |
| Function | Channeling Wealth to Real Investment: Moving wealth to real investments like building factories, buying equipment, training workers, etc.<br><br>Pricing: Capital markets send signals on where returns are high or low, driving capital accordingly.<br>High return = Attract more capital; low return = drive capital away<br><br>Liquidity:Facilitates converting investments into cash for consumption or reinvestment.<br><br>Managing Risk: Not just reducing risk but optimizing risk to suit one's risk tolerance or aversion. E.g., hedging risk through various financial instruments. |
| Securities Markets | • Equity Securities: Ownership in a company, like stocks. Investors can earn dividends and benefit from stock price increases. They are paid after debt holders if the issuer is liquidate<br><br>• Debt Securities: Loans to an issuer, such as bonds. Investors receive interest and principal repayment. They are paid before equity holders if the issuer is liquidated.<br><br>• Hybrid Securities: Mix of debt and equity (e.g., convertible bonds). |
| Derivatives Markets | Categories: Futures & forwards, options, swaps.<br><br>Derivatives allow risk-sharing and are often used for hedging or speculation. Contracts based on the value of underlying assets. Used for risk management or speculation. Gains and losses are balanced among participants (zero-sum game). |
| Currency Markets | Trading in national currencies<br>Primarily facilitates cross-border trade, investment, and changing capital forms. |
| Insurance Markets | Securitization of insurance risks (e.g., catastrophe bonds).<br>Similar to derivatives in that they optimize risk-sharing. |

# Financial Markets

14 July 2020 18:00

| Category | Details |
|---|---|
| **Capital and Money Markets** | Capital Markets: Long-term funding through debt or equity; includes activities like raising capital.<br>Money Markets: Short-term funding instruments, typically repaid within a year. |
| **Debt** | Forms of Debt: Loans (negotiated) and bonds (transferable securities).<br>Perpetual Debt: Debt instruments with no final repayment date, providing perpetual interest payments.<br>Government Borrowing: Governments use debt (mainly bonds) for funding, often borrowing in foreign currencies if domestic markets are underdeveloped. |
| **Types of Derivatives** | Swaps: Contracts to exchange cash flows based on underlying assets.<br>Forwards: Contracts to buy/sell assets at a future date at a price determined today.<br>Options: Contracts giving the right (but not obligation) to buy/sell assets at a predetermined price. |
| **Exchange vs OTC** | Exchanges provide centralized trading with transparency<br>OTC trading involves private negotiations without a centralized platform. |
| **Types of Markets** | Primary Market: A market where new financial instruments are issued for the first time; public offerings are done through a prospectus.<br>Secondary Market: A place where primary market instruments, once issued, are bought and sold. Provides an easy way to buy or sell financial assets.<br>Stock Markets: Organized trading of stocks through exchanges, where orders are matched based on quoted price, time, quantity, and type.<br>Bond Markets: Markets for issuing and trading bonds and other debt instruments. Includes government, municipal, and corporate bonds.<br>Derivatives Markets: Markets for trading derivative instruments like futures and options.<br>Money Markets: Markets for short-term debt instruments, such as certificates of deposit and commercial paper.<br>Forex Markets: Global decentralized market for trading currencies, determining exchange rates.<br>Commodity Markets: Physical or virtual marketplaces for buying, selling, and trading raw or primary products. |

# Financial Intermediaries

07 September 2024 11:24

| Aspect | Role | Description | Usage/Function |
|---|---|---|---|
| Investment Banks(GoldMan Sachs) | Financial Advisor and Capital Raiser | Investment banks play a critical role in the issuance of new securities (such as IPOs or bond issuances) and act as underwriters, helping firms raise capital. They also provide advisory services related to mergers, acquisitions, and other corporate finance activities | • Underwriting: Investment banks assess the risk and help companies sell their new securities to the public by buying these securities and reselling them to investors.<br>• Advisory: They advise companies on major financial decisions, such as going public, raising debt, or selling assets |
| Commercial Banks(JPMC) | Loan Provider and Trader | Offer loans, deposits, and may also engage in trading and underwriting. | Provide loans, handle deposits, engage in trading. |
| Broker(Robinhood) | Trade Executor | Executes trades on behalf of clients without capital risk. | Facilitate client trades, earn commissions. |
| Dealer(GoldMan Sachs) | Proprietary Trader | Trades securities for their own account, buying and selling to make a profit. | Provide liquidity, profit from bid-ask spread. |
| Broker-Dealer Firms(Charles Schwab) | Facilitator and Trader | Broker-dealers facilitate the buying and selling of securities for clients in secondary markets. They act as brokers (executing trades on behalf of clients) and dealers (trading on their own behalf). | Earn commissions, trade for own account. |
| Market Maker(Citadel Securities) | Liquidity Provider | Quotes bid (buy) and ask (sell) prices to ensure liquidity in the market. | Profit from bid-ask spread, maintain market efficiency. |
| Proprietary Trading(Hedge Fund) | Firm Trader | Firms trade using their own capital to profit from market movements. | Engage in short-term or long-term trading. |

# Equity

14 July 2020 11:11

| Category | Details |
|---|---|
| **Equity Definition** | Tradable securities representing ownership in a company, also known as shares or stock. |
| **Ownership Representation** | Represents a percentage of ownership in a company. Example: 10 shares represent 10% each; 100 shares represent 1% each. |
| **Types of Equity Securities** | Common Stock: Voting rights, dividends if declared.<br>Preferred Stock: Fixed dividends, fewer voting rights, if a company declares bankrupcy, the preferred stock holders are paid before common shareholders<br><br>Priority level if company gets liquidated is:<br>• Debt holders<br>• Preferred Shares<br>• Common Shares |
| **Dividends** | - Cash Dividend: Monetary payout per share.<br>- Stock Dividend: Additional shares issued.<br>- Property Dividend: Tangible assets. |
| **Important Dividend Dates** | - Declaration Date: Announcement of dividend.<br>- Record Date: Entitlement to dividend.<br>- Ex-Dividend Date: Cutoff for eligibility.<br>- Payment Date: When dividend is paid out. |
| **Common vs. Preferred Stock** | - Common Stock: Voting rights, shareholder participation.<br>- Preferred Stock: Fixed dividends, paid before common stockholders. |
| **Preferred Stock Types** | - Cumulative: Accrued dividends paid before common stock.<br>- Redeemable: Issuer can repurchase.<br>- Convertible: Can be converted to common stock.<br>- Participating: Additional dividends after common stockholders. |
| **Depositary Receipts (DRs)** | Represent foreign shares and trade independently, facilitating domestic investment in foreign stocks.<br><br>American Depository Receipts (ADR)<br>The purpose of an ADR is to facilitate the domestic trading of a foreign stock. An ADR is a receipt for a specified number of foreign shares owned by an American bank. ADRs trade like shares, either on a U.S. Exchange or Over the Counter. The owner of an ADR has voting rights and also has the right to receive any declared dividends. An example would be Infosys ADRs that are traded in NASDAQ. |
| **Exchange-Traded Funds** | Instruments tracking assets or indices, traded on stock markets like individual stocks. |
| **Private Equity** | Financing for non-public companies, filling the gap between personal capital and public markets. |
| **Stock Exchanges** | - Exchanges provide primary and secondary markets. |
| **Algorithmic Trading** | Uses algorithms for automated trading decisions and execution. High-frequency trading focuses on rapid order execution. |
| **Key Order Types** | - Market Orders: Buy/sell immediately at the best price.<br>- Limit Orders: Buy/sell at specified price or better.<br>- Stop Orders: Buy/sell when price reaches a specified level.<br>- Stop-Limit Orders: Combines stop and limit orders. |
| **Other Order Types** | - Day Orders: Expire at the end of the trading day if not filled.<br>- Good 'til Canceled (GTC): Remain until executed or canceled.<br>- Fill-or-Kill (FOK): Must be executed immediately in full or canceled.<br>- Immediate-or-Cancel (IOC): Must be executed immediately, with unfilled portions canceled. |
| **Dividend Yield** | Ratio of annual dividend per share (DPS) to share price:<br>Dividend Yield = DPS / Share Price<br>Example: DPS = USD 0.50, Share Price = USD 10.00; Dividend Yield = 5%. |
| **Earnings Per Share (EPS)** | Calculated by dividing annual net income by the number of outstanding shares:<br>EPS = Net Income / Number of Outstanding Shares<br>Example: Net Income = USD 10 million, Shares = 1 million; EPS = USD 10.00 per share. |
| **Price/Earnings (P/E) Ratio** | Relates share price to EPS:<br>P/E Ratio = Share Price / EPS<br>Example: EPS = USD 0.50, Share Price = USD 10.00; P/E Ratio = 20 times. |

# Fixed Income

14 July 2020 11:12

| | |
|---|---|
| **What Is a Bond?** | A debt instrument representing money borrowed by the issuer, with repayment of principal and periodic interest payments. |

**Key Bond Features**

| Term | Definition |
|---|---|
| **Face Value (Par Value)** | The amount the bondholder will be repaid at maturity. Most bonds have a par value of $1,000, though this can vary. |
| **Coupon Rate** | The interest rate the bond issuer agrees to pay the bondholder, typically a percentage of the face value (e.g., 5% coupon on a $1,000 bond pays $50 annually). |
| **Coupon Payments** | The periodic interest payments made to bondholders, typically semi-annually or annually, based on the coupon rate. |
| **Maturity Date** | The date when the bond's principal or face value is repaid to the bondholder. Maturities can range from a few months to over 30 years. |
| **Issuer** | The entity that issues the bond and borrows the funds. Issuers include governments, municipalities, corporations, and international organizations. |
| **Yield** | The return a bondholder gets on the bond. Types include: |
| | • **Current Yield**: Annual coupon payment divided by the bond's current market price. |
| | • **Yield to Maturity (YTM)**: Total return expected if the bond is held to maturity, considering coupon payments and price difference. |
| **Credit Rating** | Bonds are rated by agencies (e.g., Moody's, S&P, Fitch) based on issuer creditworthiness. Higher-rated bonds (e.g., AAA) are safer, while lower-rated bonds offer higher yields but carry more risk. |
| **Market Price** | Bonds can be traded in the secondary market, and their price can fluctuate above or below face value based on factors like interest rates, inflation, and the issuer's creditworthiness. |

**Types of Bonds**

| Type of Bond | Definition | Examples/Types | Features |
|---|---|---|---|
| **Government Bonds** | Bonds issued by national governments to finance spending. Considered one of the safest investments, especially in stable countries. | U.S. Treasury Bonds, Gilts (UK), Sovereign Bonds | Low risk, used as a benchmark for risk-free rates. |
| **Municipal Bonds (Muni Bonds)** | Bonds issued by local governments or agencies to fund public projects like schools, roads, utilities. | General Obligation Bonds, Revenue Bonds | Interest income often tax-exempt; safer than corporate bonds, riskier than government bonds. |
| **Corporate Bonds** | Bonds issued by corporations to raise capital for business activities such as expansion or research. | Investment-Grade Bonds, High-Yield (Junk) Bonds | Higher yields than government bonds, credit quality varies with company's financial health. |
| **Zero-Coupon Bonds** | Bonds issued at a deep discount with no regular interest payments; mature at face value. The difference between purchase price and face value is the return. | U.S. Treasury STRIPS | Ideal for lump-sum payout at maturity; more sensitive to interest rate changes. |
| **Convertible Bonds** | Corporate bonds that can be converted into a set number of shares of the issuing company's stock. | - | Potential for capital appreciation, lower coupon rate than regular corporate bonds. |
| **Inflation-Linked Bonds** | Bonds with principal or interest payments indexed to inflation, protecting purchasing power. | U.S. TIPS (Treasury Inflation-Protected Securities) | Attractive to investors concerned about inflation eroding fixed-income returns. |
| **Perpetual Bonds** | Bonds with no maturity date, meaning they pay interest indefinitely. | - | Issuer pays interest forever; higher coupon rates due to perpetual nature. |
| **Foreign Bonds** | Bonds issued in a country by a foreign entity in the domestic currency of that country. | Yankee Bonds (U.S.), Samurai Bonds (Japan) | Currency risk for investors; payments in the issuer's domestic currency. |
| **Callable Bonds** | Bonds that allow the issuer to redeem the bond before the maturity date, usually at a premium. | - | Issuers may call bonds if interest rates drop; offer higher yields to compensate for call risk. |

| | |
|---|---|
| **Price vs. Yield** | Yield has an inverse relationship with bond price. If bond price rises, yield falls, and vice versa. |
| **Types of Fixed Income Instruments** | - Bonds: Loans by governments or corporates, with interest paid as the coupon rate.<br><br>- Par/Face/Principal Value: Amount returned at maturity; prices fluctuate with interest rates.<br><br>- Coupon Rate: Annual interest rate, typically paid every six months; zero-coupon bonds do not pay periodic interest. |

| | |
|---|---|
| | - Maturity Period/Term to Maturity: Date on which principal value is paid; ranges from 1 month to over 40 years. |
| **Classification of Bonds** | - Central Government Bonds (Treasuries): Highly liquid, low-risk.<br><br>- Treasury Bills (T-bills): Short-term loans maturing within a year.<br><br>- Treasury Notes: Bonds maturing within one to ten years.<br><br>- Treasury Bonds: Bonds maturing after ten years.<br><br>- Municipal Bonds: Issued by local governments.<br><br>- Corporate Bonds: Issued by corporations.<br><br>- Mortgage Bonds: Secured by property.<br><br>- Collateral Trust Bonds: Secured by financial assets.<br><br>- Zero-Coupon Bonds: Issued at a discount, redeemed at face value; difference between purchase price and face value is the return. |
| **Default Risk** | - Refers to the risk of not receiving the promised money in full or on time. |
| **Interest Rate Risk** | - Comprises **Price/Market Risk** and **Reinvestment Risk**. |
| **Purchasing Power Risk (Inflation/Deflation)** | - Uncertainty about what the bondholder can buy with future payments.<br>Inflation decreases purchasing power; Deflation increases it, benefiting bondholders.<br>Longer maturities are more exposed to this risk. |
| **Liquidity Risk** | - The ease of converting an investment into cash.<br>- Government securities are highly liquid, while less liquid investments offer higher returns but increased risk. |
| **Sources of Return** | C**ash Flows** (coupons and principal repayments)<br>**Reinvestment Income** (return on cash reinvested)<br>**Price Changes** (driven by yield changes). |

# Bond Calculations

14 July 2020 11:14

**Primary characteristics of a bond :**

- Price of the bond
- Rate of return
- Maturity period
- Price

Bond is nothing but a series of cash flow. When you buy a bond you have negative cash flow, and when you get the coupons, they can be equalled as positive cash flows. Thus we need to discount the cash flows with the interest rates prevalent. The rate used should be the opportunity cost of capital, i.e. the rate of a similar risk – same maturity bond.

We will first try to understand the various rates that are normally mentioned:

- **Spot Rate**: Spot rate is the interest rate on an investment starting today and ending after some specified day. Thus 3 year spot rate from today means the rate of return on an instrument with a three-year maturity and starting from today.
- **Forward Rate:** A forward rate is an interest rate contracted today on an investment that will be initiated after some time in future. In other words it is spot rate in future. Forward rates and spot rates are interlinked.

| Years | Spot rates |
|---|---|
| 1 | 5% |
| 2 | 6% |

- Here Row 1 shows that if one invests 100 Rs. today then after 1 year it will be 105 Rs.
- Also Row 2 shows that if one invests 100 Rs. today, after 2 years it will become 100* (1.06)^2 = 112.36 Rs.

1-year forward rate from next year: For that, you will have to see that one year forward rate (fr1) has to satisfy following equation:

**Investing in 100 Rs today for one year and then re-investing the payments received after the first year for another year = Investing the 100 Rs today for two years.**

i.e. sr1 = Spot Rate for 1 year
sr2 = Spot Rate for two years
1fr1 = Forward rate for 1 year starting one year from now

Therefore, 100 * (1+ sr1 )(1+ 1fr1) = 100(1+ sr2)^2

100(1.05) (1+fr11) = 100(1.06)^2

1fr1 = (112.36/105) – 1 = 7%

Now why have we told you about the concept of forward rates and spot rates? For one thing, you will have to know which rate to use while discounting the cash flows of a bond. Whenever we calculate the NPV we use the spot rate. The reason is because we want to discount the cash flow to the present date. Hence any rate that we use should identify the rate from present date. So what will we do if we just have forward rates? In that case we will calculate the spot rates based on the forward rates. Actually the price of a bond fluctuates. This is so because the total amount of return from a bond is fixed. Thus as the spot rate of the economy changes the present value of the future amounts would also vary. To make it less confusing let's look at this example.

Suppose you have a bond with a face value of 100 $ and will give a 10 % coupon for next three years. So the pay offs are:

| | Returns | Investment | Yearly Returns |
|---|---|---|---|
| Present Yr | 0 | 100 | -100 |
| First Yr | 10 | 0 | 10 |
| Seconds Yr | 10 | 0 | 10 |
| Third Yr | 110 | 0 | 110 |

$$\textbf{Bond Price} = \sum_{i=1}^{n} \frac{C}{(1+r)^n} + \frac{F}{(1+r)^n}$$

where C = Periodic coupon payment,

- F = Face / Par value of bond,
- r = Yield to maturity (YTM) and
- n = No. of periods till maturity

So you can see that the total investment is 100 $ and the returns are 130$. The 10$ in the third year does not have the same value as 10$ in the first year. So the actual returns will have to be the discounted values of the cash flows to the present year.

Let us assume that the spot rate of a bond having similar risk is 10% in first 2 years and 5 % in third year. Thus after calculating NPV the table looks like

| Year | Returns | Investment | NPV | Total yearly Return |
|---|---|---|---|---|
| Present Year | 0 | 100$ | 100$ | -100$ |
| First year | 10$ | 0 | 9.09$ | +9.09$ |
| Second Year | 10$ | 0 | 8.26$ | +8.26$ |
| Third Year | 110$ | 0 | 95.02$ | +95.02$ |
| | Total Inflow (Add 2,3,4 rows) | | | **+112.37$** |

**9.09**=10/(1+0.1)
**8.26**=10/(1+0.1)^2
**95.02**=10/(1+.05)^3+(100/(1+.05)^3)

**Yield**

The yield of a bond is, roughly speaking, the return on the bond i.e. overall return expected from a bond, considering both the coupon payments and any capital gains or losses if held until maturity.The yield is expressed as the annual rate based on the par value.

There are several types of yield:

- **Nominal** yield is the same as the coupon rate. It is the rate of income that you receive based on the par values of the bond. Supposing you buy a bond with par value of 1000 Rs. and a coupon of 100 Rs Annually. Then the nominal yield is 100/1000 = 10 %.
- **Current** yield is a measure of the return on the bond based on the current price. Since the current price of a bond fluctuates the current yield of a bond also fluctuates. If the par value of a bond is 1000 Rs with a coupon of 120 Rs (i.e. 12%), and the current price is 1200 Rs then the current yield is 10%.
- **Yield to maturity** is the overall return on the bond if it is held to maturity. It reflects all the interest payments that are available through maturity and the principal that will be repaid, and **assumes that all coupon payments will be reinvested at the current yield on the bond**. This is the most valuable measure of yield because it reflects the total income that you can receive. If you purchase the bond at a discount, yield to maturity will reflect the fact that at maturity you will have additional income because of the difference between the price paid and the principal returned. For example, if you purchase a $1,000 par value bond for $800, you have $200 extra in income at maturity. The calculation is slightly different if the bond is purchased at a premium, but it takes into account the extra amount paid for the bond.

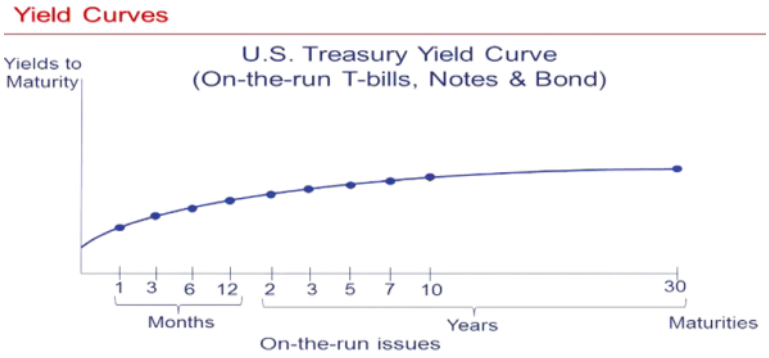
| | Returns | Investment | Total yearly Return |
|---|---|---|---|
| Present Year | 0 | 100$ | -100$ |
| First year | 10$ | 0 | 10$ |
| Second Year | 10$ | 0 | 10$ |
| Third Year | 110$ | 0 | 110$ |

**Nominal Yield:**
The nominal yield is the coupon rate, which is
Coupon value/ original price of bond
= 10/100 = 10 %
**Current Yield**
Current yield is:
Coupon value/ Current market price
=10/112.37 = 8.89%
**Yield to Maturity**
YTM for the bond is the rate at which the NPV of the bond is zero. Thus the YTM is when "r" when:
112.37 = 10/(1+r) + 10/(1+r)^2 + 110/(1+r)^3
Hence YTM is 5.4%

- Normal Curve: Upward sloping; longer-term yields higher than short-term.
- Inverted Curve: Short-term rates higher than long-term rates; potential recession signal.
- Flat or Humped Curve: Similar short-term and long-term rates, or medium-term rates differing from short/long-term.

Yield Curve: graphical representation of interest rates (yields) for securities of the same type but with different maturities. It shows the relationship

between the interest rates and the time to maturity of those securities



# FX

07 September 2024 12:47

| Aspect | Explanation |
|---|---|
| **Definition of FX Trading** | FX trading involves buying and selling currencies in the foreign exchange market. It facilitates international trade, investments, and finance by allowing currency conversion. |
| **Key Participants** | 1. **Banks/Financial Institutions**: Conduct large volume trades on behalf of clients or themselves.<br>2. Corporations: Exchange currencies for international trade.<br>3. Governments/Central Banks: Manage reserves and rates.<br>4. Investors/Hedge Funds: Speculate on currency movements.<br>5. Retail Traders: Trade small amounts via brokers. |
| **Currency Pairs** | Currencies are traded in pairs (e.g., EUR/USD). The price indicates how much of the quote currency (USD) is needed to buy one unit of the base currency (EUR). Example: EUR/USD = 1.15 means 1 EUR = 1.15 USD.<br><br>Buying EUR/USD: A trader buys the EUR/USD pair if they believe the Euro will strengthen against the U.S. Dollar (the exchange rate will rise).<br>Selling EUR/USD: A trader sells the EUR/USD pair if they believe the Euro will weaken against the U.S. Dollar (the exchange rate will fall). |
| **Types of FX Markets** | 1. Spot Market: Immediate currency exchange at current market prices.<br>2. Forward Market: Currency exchanged at a future date, agreed rate (used for hedging).<br>3. Futures Market: Standardized contracts traded on exchanges for future delivery.<br>4. Options Market: Gives the right (not obligation) to exchange currency at a future date/price. |
| **Trading Mechanism** | FX trading is OTC (over-the-counter), meaning trades happen directly between participants (banks, brokers, investors) via electronic platforms or internal systems. |
| **Trading Hours** | FX market operates 24 hours during the working week, spanning multiple time zones (London, New York, Tokyo, Sydney), ensuring continuous liquidity. |
| **Bid/Ask Spread** | 1. Bid Price: Price at which traders are willing to buy a currency.<br>2. Ask Price: Price at which traders are willing to sell a currency.<br>3. Spread: Difference between bid and ask price, representing transaction costs. A narrower spread typically indicates a more liquid market, while a wider spread can indicate lower liquidity or higher transaction costs. |
| **FX Risks** | 1. Currency Risk: Fluctuations in exchange rates.<br>2. Interest Rate Risk: Changing interest rates affect currency values.<br>3. Credit Risk: Default risk in OTC transactions.<br>4. Liquidity Risk: Difficulty trading without affecting price. |

# Hybrids

14 July 2020 11:15

| Aspect | Description |
|---|---|
| **Definition** | Financial instruments that combine features of both debt and equity. |
| **Types of Hybrids** | 1. Convertible Bonds: Can be converted into shares.<br>2. Convertible Preferred Shares: Fixed dividends and convertible to common stock.<br>3. Perpetual Bonds: Indefinite interest payments, no maturity.<br>4. Preferred Stock: Fixed dividends, priority over common stock.<br>5. Subordinated Debt: Lower ranking debt, riskier.<br>6. Contingent Convertible Bonds (CoCos): Convert to equity on trigger events. |
| **Benefits** | - Income generation through regular payments.<br>- Potential for capital gains via conversion to equity.<br>- Diversification of risk/reward profile. |
| **Risks** | - Interest rate sensitivity affecting prices.<br>- Credit risk, as they may be repaid last in distress.<br>- Market risk related to underlying stock performance. |
| **Purpose** | - Capital Flexibility: Raise capital with less debt impact.<br>- Attractive to Investors: Fixed income with equity upside.<br>- Balance Sheet Benefits: Improve debt-to-equity ratio without immediate dilution. |
| **Example** | Convertible bonds provide regular interest and the option to convert to stock, offering the potential for appreciation if the stock performs well. |

# Derivatives

14 July 2020 17:26

A derivative is a product whose value is derived from the value of an underlying asset, index or reference rate. The underlying asset can be equity, foreign exchange, commodity or any other item.

| Feature | Description |
|---|---|
| **Underlying Asset** | The asset, index, or benchmark that determines the derivative's value (e.g., stocks, bonds, commodities). |
| **Leverage** | Derivatives allow traders to control a larger position with a smaller amount of capital, magnifying both gains and losses. |
| **Contractual Agreement** | Derivatives are agreements between parties that specify terms, including the underlying asset, price, expiration date, and settlement method. |
| **Market Complexity** | Derivatives can be complex and involve various factors, including interest rates, volatility, and time until expiration. |
| **Regulatory Framework** | Derivatives trading is subject to regulation to ensure transparency, reduce systemic risk, and protect investors. |

| Type | Description |
|---|---|
| **Forwards** | Customized contracts between two parties to buy or sell an asset at a specified future date and price. Not traded on exchanges and have counterparty risk. |
| **Futures** | Standardized contracts to buy or sell an asset at a predetermined price on a specific date in the future. Traded on exchanges, which reduces counterparty risk. |
| **Options** | Contracts that give the holder the right, but not the obligation, to buy or sell an asset at a specified price before a certain date. Types include call and put options. |
| **Swaps** | Agreements between parties to exchange cash flows or other financial instruments. Common types include interest rate swaps and currency swaps. |
| **Exotic Derivatives** | More complex derivatives with features not found in standard derivatives, often tailored to specific needs (e.g., Asian options, barrier options). |

# Options

10 October 2024 09:55

An option is a contract that gives the buyer the right, but not the obligation, to buy or sell shares of the underlying security at a specific price (the strike price) on or before a specified date (the expiration date). The value of options depends on the value of an underlying investment, which can be a stock, index, currency, commodity, or other securities.

Key Features of Options:

- Underlying Asset: Options are based on various underlying assets.
- Strike Price: The price at which the holder can buy (call option) or sell (put option) the underlying asset.
- Expiration Date: The date by which the option must be exercised.
- Premium: The price paid for the option, which is the cost to purchase the option contract.

| Type | Description |
|---|---|
| Call Option | Gives the holder the right to buy the underlying asset at the strike price before the expiration date. Typically used when the investor expects the asset price to rise. |
| Put Option | Provides the holder with the right to sell shares of the underlying stock at the strike price at the maturity date. Typically used when the investor expects the asset price to fall. |
| American Options | Can be exercised at any time before or on the expiration date. |
| European Options | Can only be exercised on the expiration date itself. |
| Exotic Options | More complex options with unique features (e.g., Asian options, barrier options) that may depend on certain conditions or averages over time. |

| Type | Description |
|---|---|
| Stock Options | A contract that guarantees the investor the right to buy or sell shares of the underlying stock at a fixed price prior to a certain date. The buyer is known as the holder, and the seller is called the writer. If the option contract is exercised, the writer must fulfill the terms of the contract. |
| Index Options | A contract that guarantees the investor the right to buy or sell the underlying index at a fixed price prior to a certain date. Settlement is always on a cash basis. |

| **Strategy** | **Description** | **Market Outlook** |
|---|---|---|
| Covered Call | Holding a long position in an asset and selling call options on the same asset to generate income. | Neutral to moderately bullish. |
| Protective Put | Buying a put option while holding a long position in the underlying asset to limit potential losses. | Bullish (but seeks downside protection). |
| Straddle | Buying both a call and a put option with the same strike price and expiration date, profiting from large price movements in either direction. | Expecting high volatility. |
| Strangle | Buying a call and a put option with different strike prices but the same expiration date, similar to a straddle but usually cheaper. | Expecting high volatility. |
| Bull Call Spread | Buying a call option at a lower strike price and selling another call option at a higher strike price to limit potential losses. | Bullish (moderate upside). |
| Bear Put Spread | Buying a put option at a higher strike price and selling another put option at a lower strike price. | Bearish (moderate downside). |
| Iron Condor | Selling a bear call spread and a bull put spread simultaneously to profit from low volatility in the underlying asset. | Neutral (expecting low volatility). |
| Calendar Spread | Buying and selling options with the same strike price but different expiration dates, profiting from time decay and volatility differences. | Expecting moderate volatility. |

# Forwards and Futures

14 July 2020 17:41

Futures: A futures contract is a standardized, transferable, exchange-traded contract that requires delivery of a commodity, bond, currency, or stock index, at a specified price, on a specified future date. Unlike options, futures contracts convey an obligation to buy. Futures are formal in nature and regulated.

Forward: The mechanism of forward contract is the same as the ones of a futures contract. The only difference is that the forward contract is not traded on an exchange. A forward contract is a direct agreement between two investors

| Aspect | Forwards | Futures |
|---|---|---|
| **Customization** | Highly customizable | Standardized |
| **Trading Venue** | Over-the-counter (OTC) | Traded on exchanges |
| **Settlement** | Settled at expiration | Can be settled physically or in cash |
| **Counterparty Risk** | Higher due to lack of regulation | Lower due to exchange clearinghouses |
| **Margin Requirements** | None (unless otherwise agreed) | Requires margin deposits |
| **Regulation** | Less regulated | Highly regulated |

# Swaps

14 July 2020 17:49

Swaps(OTC) are agreements between at least two counter-parties to exchange cash flows in the future according to a pre-specified formula. They can therefore be regarded as portfolios of forward contracts. The most common one is an agreement on the exchange of a fixed rate for a floating rate contract.

**Interest Rate Swaps**

Interest rate swap is the exchange of one set of cash flows for another. A pre-set index, notional amount and set of dates of exchange determine each set of cash flows. The most common type of interest rate swap is the exchange of fixed rate flows for floating rate flows.

**Example**

CTS comes up with a deal with ICICI bank in which CTS pays ICICI bank an Interest of 7% fixed on 100 crore rupees. In turn ICICI promises to pay CTS a monthly interest of inflation rate + 1% on 100 crore rupees. Thus assuming that for 6 months the inflation rate was 5%, 5.25%, 6%, 6.75%, 5.5%, 5.5% then the pays offs would be as follows:

| | Month I | Month 2 | Month 3 | Month 4 | Month 5 | Month 6 | Total |
|---|---|---|---|---|---|---|---|
| Inflation rate | 5 | 5.25 | 6 | 6.75 | 5.5 | 5.5 | |
| ICICI pays | 6 | 6.25 | 7 | 7.75 | 6.5 | 6.5 | 40 |
| CTS pays | 7 | 7 | 7 | 7 | 7 | 7 | 42 |

So CTS pays 42 crores whereas ICICI pays 40 crores. Do you think CTS went in loss? Well actually it has, but if you see that the loss of 2crore has helped CTS transfer the risk of inflation rate volatility. Thus ICICI has in a way bought the inflation rate risk for Rs 3 crores.

In case the inflation rate would have increased to 8 or 9 % then ICICI would have had to pay the additional money to CTS, and CTS would have been completely protected from the high inflation rates.

*Note:* In case you haven't noticed, the 100 crores of principal amount have NOT been exchanged. The only exchange is of the interests and the principal is just a notional amount.

**Currency Swaps**

Currency swap is a swap contract in which two counterparties agree to exchange principal and interest denominated in different currencies based on an agreed-upon currency exchange rate. In an interest rate swap, we were concerned exclusively with the exchange of cash flows relating to the interest payments on the designated notional amount. However, there was no exchange of notional at the inception of the contract. The notional amount was the same for both sides of the currency and it was delineated in the same currency. Principal exchange is redundant.

**Equity Swaps**

Equity swaps are exchanges of cash flows in which at least one of the indices is an equity index. An equity index is a measure of the performance of an individual stock or a basket of stocks. Common equity indices with which the general investor is probably familiar include the Standard & Poor's 500 Index, the Dow Jones Industrial Average or the Toronto Stock Exchange Index.

**Credit derivatives**

It is a contract between two parties that allows for the use of a derivative instrument to transfer credit risk from one party to another. The party transferring risk away has to pay a fee to the party that will take the risk.

**Asset Based Securities**

A 1000 people take loans from ICICI bank with a combined value of 100 crore Rs. ICICI bank decides to get money based on these loans. So it groups together the loans worth 100 crores Rs. and starts selling them as securities. Now the investors buy the securities. The dividends that they get are nothing but the interest payments form these investors. Of course some of the investors may not pay the interests and some may default altogether. This will fluctuate the price of the ABS. Also the change in the interest rates will change the price of the underlying loans, thus changing the value of the assets.

# Facilitation Institutions

30 July 2020 09:01

| Entity | Explanation | Analogy | Example |
|---|---|---|---|
| **Custodian** | Safely holds securities and handles income collection, settlements, and reporting. | Bank locker for stocks | Holds your Apple shares and collects dividends. |
| **Exchange** | Marketplace where securities are traded with standard rules and transparency. | Supermarket | NYSE or Nasdaq brings buyers and sellers together. |
| **Clearing Firm** | Ensures trade settlements between buyers and sellers, reducing risks. | Referee in a trade | Guarantees you get your shares and the seller gets paid. |
| **Clearing Bank** | Handles money transfers for clearing firms to settle trades. | Payment gateway | Moves your payment to the seller when you buy shares. |
| **Depository Trust Company (DTC)** | ensures ownership changes hands smoothly through electronic settlement. | | Transfers Google shares digitally from seller to buyer. |

# Buy Side and Sell Side

30 July 2020 09:12

**Buy side** is the side of Wall Street comprising the investing institutions such as mutual funds, pension funds and insurance firms, that tend to buy large portions of securities for money-management purposes. A buy-side analyst typically works in a non-brokerage firm (i.e. mutual fund or pension fund) and provides research and recommendations exclusively for the benefit of the company's own money managers. A buy-side analyst would only be interested in analysing whether the investment is suitable for the firm's investment strategy and portfolio. Buy-side recommendations are not available to anyone outside the firm. Buy-side analysts often source research from sell-side analysts, and then use this information as a base for their own research.

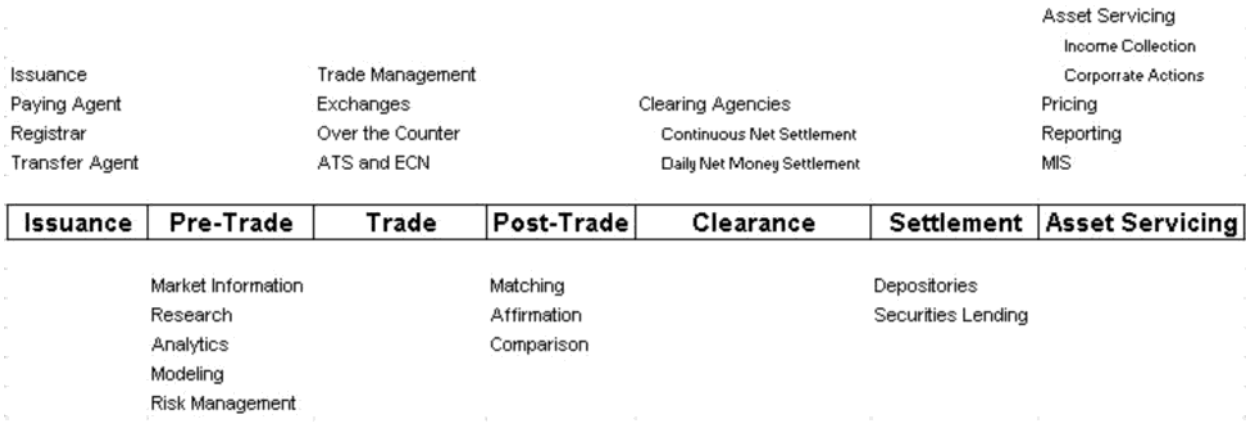
The **sell side** entities comprise brokerage firms, investment bankers and research departments that sell securities and make recommendations for brokerage firm's customers. These entities provide recommendations for upgrades, downgrades, target prices and opinions to the public market. Sell-side analysts are employed by brokerage firms, and the sell-side recommendations are meant for the general investing public. A sell-side analyst's focus when analysing possible investments is to see whether the investment should be recommended to the firm's clients. Sell-side analysts are those who issue the often-heard recommendations of "strong buy", "outperform", "neutral" or "sell". These recommendations help clients make decisions to buy and / or sell certain stocks.

Sell side firms also directly interact with the exchange by placing orders in securities. It’s the responsibility of the sell side to ensure the best execution price for an order from the buy side. They also help firms raise money from the market by creating new securities.

Thus a buy side firm like a mutual fund would design a portfolio of securities in which investment needs to be made and contact the sell side brokerage firm that would in turn place orders on the exchange on behalf of buy side. If an institutional portfolio manager changes jobs and becomes a registered representative, he or she has moved from the buy side to the sell side.

# CTS:Trade Life Cycle

09 February 2021 11:43




| Stage | Description |
|---|---|
| Issuance | • **Overview**: The trade life cycle starts with the creation of a financial instrument (e.g., stocks, bonds). For example, a company raises capital through an Initial Public Offering (IPO), where shares are sold to the public for the first time, or a governm ent may issue bonds via auctions.<br>• **Key Insight**: Post issuance, these securities may be listed on exchanges, such as the NYSE, for secondary market trading. Some securities, like government bonds, may not be listed and may be traded Over the Counter (OTC). |
| Pre-Trade | Pre-Trade Analysis: Investors or firms conduct research to analyze the security and evaluate its impact on their portfolio. This stage involves:<br><br>• Risk management: Assessing the liquidity and credit risk of the security.<br>• Portfolio management: Modeling how the trade affects the investor's entire portfolio.<br>• Client Onboarding: A key component, which involves establishing a relationship between the institution and the client, ensuring the client is in the system and legally authorized to trade. Relevant documentation, such as ISDA agreements (for OTC derivatives), is prepared.<br>• Compliance and Regulations: Ensure that trading complies with laws (like MiFID in the EU or Dodd-Frank in the US).<br>• Collateral Management: Collateral is often required to mitigate risks.<br>• Controls and Legal Agreements: Before any trades, institutions must put proper controls and agreements in place, especially for OTC derivatives. |
| Trade Execution | Order Placement: After pre-trade analysis, investors place orders to buy or sell securities. This is typically done through brokers or directly on an exchange.<br>Types of orders include:<br><br>Market Order: Executes immediately at the current market price.<br>Limit Order: Executes only at a specified price or better.<br>Stop Order: Executes when the security reaches a certain price.<br><br>Motivation for Trades: Clients may trade due to various motivations, such as a need for cash, hedging, or portfolio diversification.<br>Trade Communication: Clients and institutions communicate during execution, and trades may occur on exchanges or other venues, including OTC markets. |
| Post-Trade, Clearing, and Settlement | This stage involves multiple processes that ensure the trade is accurate, both sides agree on the terms, and the assets and cash are transferred between parties.<br><br>4.1 Clearing<br>Matching & Confirming: After the trade, counterparties need to confirm that the details match. This is done by the exchange or clearing house.<br>Trade Capture & Enrichment:<br>Trade Capture: Records basic information, such as asset type, price, quantity, and date.<br>Trade Enrichment: Adds detailed information, such as value dates, securities identifiers, and counterparty information.<br>Trade Validation: The trade data is validated to check for errors.<br>Affirmation & Confirmation: Direct participants (e.g., institutions) confirm the trade, and indirect participants (e.g., clients) affirm the trade.<br>Trade Reporting: Transactions are reported to regulators or appropriate reporting mechanisms.<br>Settlement Instructions: Final preparation of instructions to settle the trade is made.<br>4.2 Settlement<br>Overview: Settlement is the final exchange of securities for cash. It involves the actual transfer of ownership of the asset and the payment of cash. |

| | |
|---|---|
| | Settlement methods:<br><br>Delivery vs Payment (DVP): Cash is exchanged only when the security is delivered. This mitigates risk by ensuring both delivery and payment happen simultaneously.<br>Free of Payment (FOP): Delivery of the security and payment occur separately. This method is riskier since one party may not fulfill their end of the bargain.<br>Dates in the clearing and settlement process:<br><br>Trade Date: The date the agreement to trade is made.<br>Value Date: The contractual date on which cash and securities should be exchanged.<br>Settlement Date: The actual date on which the trade is settled. If this differs from the value date, a settlement failure occurs. |
| Asset Servicing & Ongoing Position and Risk Management | Ongoing Monitoring: Even after the trade is settled, there is ongoing position and risk management, which is critical for institutions to monitor their entire portfolio. Key tasks include:<br>Corporate actions: Managing events such as dividend distribution or stock splits.<br>P&L Calculation: Regular measurement of the profit and loss from trades.<br>Risk Management: Continuous monitoring of risk exposure, counterparty risk, and market sensitivities.<br>Reconciliation: Ensuring that the trade details recorded by the firm match those held by the counterparty and clearinghouses.<br>New insights from Set 2:<br>Position Management: The institution manages multiple positions within its trading book, ensuring proper risk assessment across all trades.<br>Regulatory Reporting: Institutions may have ongoing regulatory requirements, reporting to bodies like the SEC or CFTC. |

# Miscellaneous Points

13 February 2019 00:19

**1) Stock Exchange:** An entity that controls the business of buying and selling of securities. Stock Exchange can be regional or national exchanges.
**Example**: NASDAQ – USA, NSE – India etc.

**2) Stock/Share/Equity:** Total capital of a company is divided into equal units; each unit is termed as share/equity/stock. Stock also represents a part of ownership of a company.

**3) Face value of a Share:** The amount or value (used during buying or selling) allotted to a share by the company.

**4) Issue Price:** The price of a company's shares at which they are available in the market. When these shares are traded in the market the price may be below or above the issue price.

**5) Initial Public Offering (IPO):** This is nothing but selling the securities or shares of a company to the public for the first time in the market.

**6) Market Capitalization:** The financial value of a company is calculated by multiplying the share price with number of shares which is termed as Market Capitalization.
**Example**: Suppose a Company X has 100 shares. The current market price of each share is $50. Then the market capitalization of the Company X is $5000.

**7) Security Market:** Security market is a place where buyers and sellers of securities (bonds, debentures, stocks etc.,) do their transactions of buying and selling the securities.

**8) SEBI (Security and Exchange Board of India):** An authority that makes sure whether the buyers and sellers behave in a proper way in the market. So that they get their desired profits. There are different security and exchange boards/commissions as per the country.

**9) Dividend on share:** Dividend is a percentage of the value of a share, which a company returns to its share holders from its annual profits.

**10) Bid Price:** Bid Price is the rate at which the buyer is ready to buy the stock.

**11) Ask Price:** This is the price at which the seller wants to sell his stock.

**12) Futures:** A future contract is an agreement between the buyer and the seller in which the stock of future delivery is transacted at a particular price.
**For example**, if you want to purchase a March future contract of XYZ Company then you have to do that at the current price available in the market. Let's say that the March futures are trading at $100 per share. By the time the contract expires (last day of the contract in March month) the price of the stock may not be the same. It may be $95 or $110. Based on these price differences investors makes profits in the markets.

**13) Options:** It is a financial contract between the buyer and seller in which the buyer has the right to buy or sell a security at a particular price on or before a particular date.
Options are of two types: Calls, and Puts.
Call means the right to buy an asset at a price within a period of time.
Put means the right to sell an asset at a price within a period of time.

**14) Portfolio:** A Portfolio is a combination of various investment assets mixed and matched for gaining profits as per an investor's goal. Items that are included in the portfolio can be shares, debentures, mutual funds etc.

**15) Depository:** An entity that holds the securities and funds of depositors in an account. The two depositories in India are National Securities Depository Limited (NSDL) and Central Depository Services Limited (CDSL).

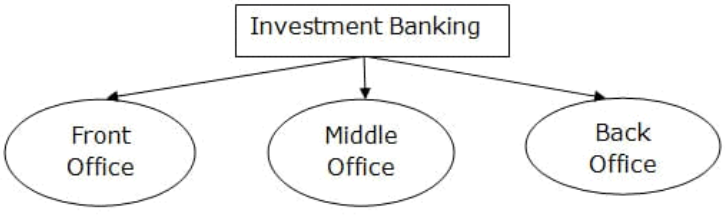
**16) Mutual Funds:** An entity that collects money from investors and invests the same in various financial instruments like shares, bonds, debentures etc.

**17) Net Asset Value (NAV):** NAV of the fund is the cumulative market value of the asset. NAV per unit is the net value of the assets divided by the number of units. Buying and selling of shares in the market are done on the basis of NAV related prices.

**18) Nifty Index:** It is a scientifically developed, 50 stock index, which shows the movement of the Indian markets. It behaves as a barometer for the Indian markets.

**19) Watch List:** A list of selected securities. It is mainly used to monitor their movement in the market regularly, closely or frequently.

## Investment Banking Organizational Structure:




**Investment banking comprises of Front Office, Middle Office, and Back Office.**

**1) Front Office:** This plays a major role in generating funds. The main areas of front office are **Investment banking, Sales & trading, and Research.**

- 'Investment Banking' helps customers in raising funds in capital markets and also suggests the companies in raising their capital.
- 'Sales & Trading' deals with buying and selling of stock (shares, bonds etc.,)
- 'Research' involves reviewing the company reports about their buy/sell ratings, company's prospects etc. This will help in providing advice to their clients in the right way.

**2) Middle Office:** This deals with 'Risk Management', 'Corporate Treasury' and 'Financial Control'.

- 'Risk Management' involves analyzing the market situations and informing the clients of the risks involved in their trades.
- 'Corporate Treasury' is responsible for the funds of Investment Banks.
- 'Financial Control' tracks the capital flow of the firm and its success.

**3) Back Office:** This includes 'Operations' and 'Technology'.

- 'Operations' checks whether the trades have been executed properly and funds transferred successfully.
- **'**Technology' supports the software, data, and systems of Investment Banks.

# Clearing and Settlement Process

15 February 2021 11:22

**Clearing** is the process of updating the accounts of the trading parties and arranging for the transfer of money and securities.

**Settlement** is the actual exchange of money and securities between the parties of a trade on the settlement date after agreeing earlier on the trade

**Process**

At the end of each trading day, concluded or locked-in trades are received. All trades concluded during a particular trading period are settled together. A multilateral netting procedure is adopted to determine the net settlement obligations (delivery/receipt positions) of Broker-Dealers . The netting procedure determines the delivery and receipt obligation of funds and securities by each member. Settlement is deemed to be complete upon declaration and release of pay-out of funds and securities.

**Players involved in the Process**

**Brokers-Dealers**: Also called members. These are the persons or the firms who are allowed to trade in an exchange.

**Clients / Custodian**: The settlement of the trades (money and securities) done by a member-broker on his own account or on behalf of his individual, corporate or institutional clients may be either through the broker-dealer himself or through a registered custodian appointed by him/client. In case the delivery/payment in respect of a transaction executed by a broker-dealer is to be given or taken by a registered custodian, then the latter has to confirm the trade done by a broker-dealer on the trading system.

· Clearing House
· Exchange

**Preparing for Clearing and Settlement**

There are 4 major steps involved in the run-up to clearing and settlement. These are:

· **Trade Matching**: This occurs when the counterparties to the trade match the details of completed trades to ensure that both sides to a trade agree in advance of the settlement. This reduces the chances of a fail due to mismatched instructions. This is usually done systematically by the exchanges and by the brokers themselves

· **Confirmation**: Confirmations are sent to investors to officially tell them what their brokers have done on their behalf. Retail investors get paper confirmations while institutional investors get electronic confirmations, though they can get paper confirmations too if they so desire. In the US, electronic confirmations are obtained through the Depository Trust Company's (DTC) automated system for securities held by the DTC. If a customer identifies an error in the confirmation, he is required to inform the broker promptly.

· **Affirmation**: Institutional investors are required to affirm the broker's confirmation; otherwise the custodian bank must let the trade fail. The affirmation acts a positive signal to the custodian to release funds or securities during settlement.

· **Netting**: A process of bringing together all of the trades made by the participants. The various trades by a participant for a specific security offset each other, thereby creating a single net debit or credit position for the participant at the end of the day for cash as well as for each security.

Netting can be of 2 types:
**Bilateral Netting**: In bilateral netting, each broker nets all of his trades with each of the other counterparties.
**Multilateral Netting**: In bilateral netting, all of each broker's trades with each of the other brokers are simultaneously netted.

| | A | B | C |
|---|---|---|---|
| A -> | 0 | 2 | 3 |
| B -> | 4 | 0 | 7 |
| C -> | 5 | 3 | 0 |

**Figure 121 : Netting**

### Case 1: No Netting:

In the absence of netting, the number of shares that have to be transferred between A, B, C are shown in the diagram below.

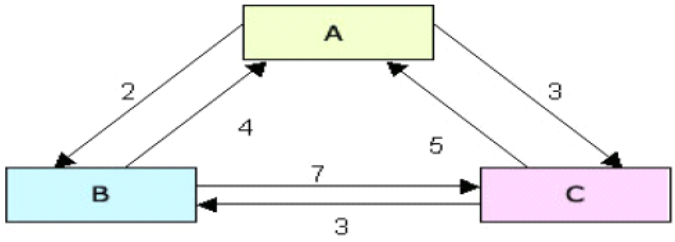



**Figure 22: No Netting**

### Case 2: Bilateral Netting:

In this case transactions between a pair brokers offset each other. For instance, instead of A transferring 2 shares to B and B transferring 4 shares to A, there will be a single transaction in which B will transfer 2 shares to A. Thus the final transactions will be as shown below.

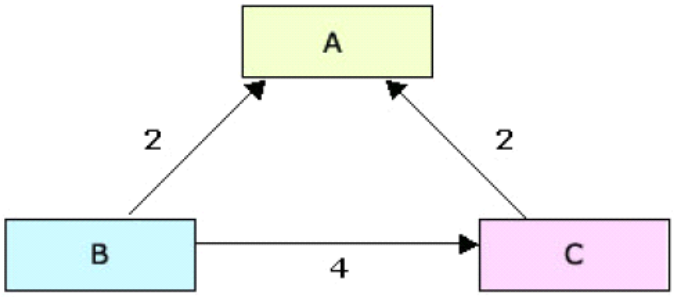



**Figure 22: Bilateral Netting**

### Case 3: Multilateral Netting:

In this case, a broker takes part in one transfer only. For instance, instead of B transferring 2 shares to A and 4 to C, B will transfer 6 shares to a body known as Clearinghouse (we shall look at this in detail later).

Trading systems based on bilateral netting are more efficient than those that don't have netting (alsoknown as trade-for-trade). Multilateral netting in turn is more efficient than bilateral netting.Now suppose that in the example above, B defaults i.e. fails to deliver the stock. Both A and C wouldsuffer from the default, as they would not get delivery of the stock. To prevent this, there are bodies known as clearinghouse who steps into the middle of a trade, becomes the counter-party to both thebuyer and the seller and guarantee completion of the transaction. It guarantees that it will complete tradeson the original terms, even if the original contra-party fails. This process of transferring obligation fromone party to the other is also known as "Novation".

# Reconciliation

18 February 2021 11:32

The process of proving that a securities trading firm's books and records are accurate is commonly known as reconciliation. An efficient trading firm seeks proof that its books and records are accurate, by comparing each component of its securities positions and cash balances with the outside world on a daily basis, as well as ensuring that its books and records reconcile internally.

**Types**

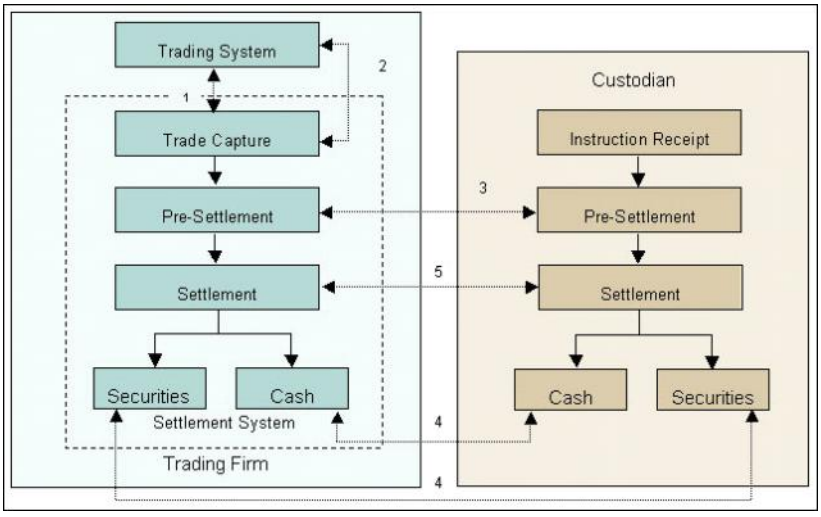
**Trade-by-Trade Reconciliation**: This is an automated reconciliation that proves that all trades captured by the traders are successfully captured within the settlement system. Not conducting this reconciliation can result in failure to process individual trades. The possibility of trades not coming into the settlement system is very much there. That's why this reconciliation is important. Its importance will increase as the settlement cycle shrinks.

**Trading Position Reconciliation**: This reconciliation is designed to prove that the trade dated security positions calculated by the trading system agree with the equivalent positions calculated by the settlement system. Not conducting this reconciliation can result in trades being executed from the incorrect trading position, where the trading system is found to be incorrect. Discrepancies can arise where trades have not been received by the settlement system, and where trade amendments and cancellations have been effected within one system but not the other. To be certain of being in a fully reconciled position, both the trade-by-trade and trading position reconciliations should be done.

**Open Trades Reconciliation**: This reconciliation aims to prove that open trades held within internal books and records are actually open at the relevant custodian, or generally have the same status at the custodian.

**Custodian Position Reconciliation**: The aim of this reconciliation is to prove that settled securities positions held within internal books and records agree with the equivalent positions as advised by the custodian. This is therefore another fundamental control and confirms to the firm that the securities that it believes are held by the custodian are actually held by the custodian.

**Settlement System Integrity Reconciliation**: This reconciliation intends to prove that the settlement system is in balance i.e. the total of securities owned is equal to the sum of the location position. If the firm employs double-entry bookkeeping, theoretically this will be true by default. Still this reconciliation must be performed frequently to prove that the system is in balance.




**Methods of Reconciliation**

The reconciliation described above can be categorized into either non-cumulative reconciliation or cumulative reconciliation.

**Non-cumulative Reconciliation**
The trade-by-trade, open trade, settlement and custody system integrity reconciliation do not involve an opening and closing balance. Rather they are 'snapshot' type reconciliation. Such reconciliation is called non-cumulative reconciliation.

**Cumulative Reconciliation**
The trading position and the custodian position reconciliation involve opening balances, movements and closing balances and are therefore called cumulative reconciliation.

# Options Explanation

23 July 2021 16:59

1. Assigning :assigning an existing owned option to the exercise

Net outcome
Cost of Buy =(1* 100)
Proceeds from sale = (2*100)
Net Proceeds=100

Strike price of Owned Option :100
Strike price of sold option: 120
Market price: 150

Net proceeds from assignment for seller of the call= (100*120)- (100 *100)+ Net Proceeds from initial sale of option
=2000+100
=2100

Net outcomeCost of Buy = (1*100)Proceeds from sale = (2*100)Net proceeds = 100Strike price of Owned Option: 100Strike price of sold option: 120Market price: 150Net proceeds from assignment for seller of the call= (100*120)-(100*100) + Net Proceeds from initial sale of option.<= 2000 = 2000 +100=2100 2100

Net proceeds from exercise for the exerciser (person who bought the option you sold)
= (Market Price *100)- (Cost of Option+(Strike Price* 100))
=15000- (200+12000)
=15000- 12200=2800

# Basic Option Strategies

12 December 2021 10:18

- **<u>Buy 2 TSLA Jan 30 calls @4(Bullish)</u>**

1. Owner has a right to buy 2x100 shares of TSLA Before expiration at price of 30
2. 2 Contracts=200 Shares
3. Options expire on Saturday following 3rd Friday of the Month
4. 30 is Strike Price
5. Calls: Right to Buy
6. 4 is Premium
7. Jan 30 is Series: Combination of expiration and Strike price

**Out of Money option**:Market Value(20)<Strike Price-> Option wont be exercised and owner will suffer a loss of 800(4X2X100).
**In the Money option**: Market Value(50)>Strike Price ->Owner will exercise the option and gain 3200(20-4=16,16x200=3200)
**At the Money option**: Market value=Strike Price

**Max Gain**: Unlimited
**Max Loss**: 4/share,
**Break Even**: when share goes to 34

- **<u>Write 10 TSLA May 40 call @3(Bearish)</u>**

Writing(Selling Calls/Naked Call): obligated to deliver/sell stock at strike Price, options are writted to generate income.

Max Gain: When market value=0, 3 Per share
Max Loss:Unlimited
Break Even: market Value:43

- **<u>Buy 1 TSLA Oct 30 put @4(Bearish)</u>**

Buying Put Options Right to Sell:

In the Money: MV<SP
Out of Money:MV>SP

Max gain: When MV=0, MG=26/share
Max Loss: 4/share
Break Even: 26 share price

# CRD

13 October 2022 08:30

**Order Life Cyle:**

Prelim: When the order is created
Pre Alert: Alerts and warnings are created
Pre-ok: Once all the alerts and warning are closed order moves to pre ok status
New: When send to trading is done
Open: when trader takes ownership
Work: either the trader has accepted the order or its in flex
Ready: order is filled
Accounting: Order is recorded in books of accounts

**Manager Workbench**- is a portfolio management tool where we can see whole portfolio at a glance and based on that we can take the action. Like Portfolio allocation. 60% equity,30% fixed income,10% cryto
A Portfolio manager cannot edit on each security manually instead he can edit on values . For eg. I want to sell 10000$ shares, then ill say -10000$ then 10000 will be deducted and automatically sell ordr will be created called as slicing

**Rebalance**- Means matching Portfolio returns with Market returns, and doing buy/sell whenever its required.
In CRD we select account and select model.
Model like high risk taking individuals ,moderate risk taking individual, conservative individual

**Navigation To Rebalnce** Window-->Manager workbench-->Accounts-->select model-->enter Account number--> press ok-
Then we will see Cur %(2.2) and Model%(1.97)

Click on 2nd last dropdown icon and click rebalance.

There will be some discrepancies and once rebalance is done all discrepancies will be resolved and sell orders will be created at one go.

Compliance- user restrictions placed for order so that they don't violate regulatory requirements or customer requirements or any other conditions put on it

Eg. cannot trade any security which is selling tobacco
For amazon I don't want 20% of Amzn exposure

If these rules are violated we get alerts

Warnings approved by portfolio manager

Alerts can be approved by algo team and assign compliance team, if compliance team is happy they will approve and then order can go ahead.

If its reject -PM has to correct the order he has to reduce the amount

If its reject -PM has to correct the order he has to reduce the amount

**Once approved order is moved to pre ok status**

**Compliance runs on whole basket.**

**--Post compliance run if order is split , orders should retain completed status**
**--Post compliance run if order is merged , each allocation should go to prelim**

**Cancel send to accounting done by trader**

Its done due to :
1. We want to revert the order
2. We want to replace it
3. Manual correction is required for the order.

Action--> cancel send to account
It will cancel and create another order in ready status